

Homework 2: EM Algorithm and Motif Finding

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```
set.seed(42)
library(patchwork)
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

An important driver of cancer is the mutation of tumor suppressor genes (TSG), among which we find genes coding for DNA repair proteins. The mutation and inactivation of such genes has the effect of increasing the frequency of mutations in a cell and its progeny. Suppose that we are investigating the inactivation of one specific TSG, and hypothesize that cells with this inactivation have a different probability of mutations than the ones without. The data we observe are sequencing results for certain positions of the genome, from a cell population at single cell resolution. We wish to find out from the data the effect of this TSG inactivation – how different are the probabilities of mutations with or without this inactivation? – as well as the proportion of cells with this inactivation in the population.

(4) TSG inactivation problem formulation

We model this problem with a two-component mixture model, corresponding to the cell groups with and without the TSG inactivation. Each component is parametrized by a mutation probability. Within a cell group, the probability of mutations are assumed to be the same at each position and the mutations are independent between positions.

More precisely, let $D = \{X^{(1)}, \dots, X^{(N)}\}$ be a set of sequences. Each $X^{(i)}$ has length L (we write $X^{(i)} = (X_1^{(i)}, \dots, X_L^{(i)})$), and $X_\ell^{(i)} \in \{0, 1\}$. In other words, we sequence N cells at L genomic positions, and represent the mutation status at each position by a binary variable, with 1 meaning mutated. Each sequence $X^{(i)}$ is generated either

- by the **mutated model** with parameter $\theta_1 \in (0, 1)$, where $P(X_\ell^{(i)} = 1) = \theta_1$, or
- by the **wild-type model** with parameter $\theta_2 \in (0, 1)$, where $P(X_\ell^{(i)} = 1) = \theta_2$.

The mutated model is used with probability λ_1 , and the wild-type model is used with probability $\lambda_2 = 1 - \lambda_1$.

We will learn maximum likelihood parameters $\lambda = (\lambda_1, \lambda_2)$ and $\theta = (\theta_1, \theta_2)$ using the EM algorithm.

(4a) The complete-data log-likelihood

Let us define binary hidden variables $Z^{(ik)}$, where $Z^{(ik)} = 1$ if and only if $X^{(i)}$ is generated from component k . We write down the complete-data log-likelihood $\log P(X, Z | \lambda, \theta)$ in terms of $X^{(i)}$, $Z^{(i)}$, λ , and θ .

$$\log(P(X, Z | \lambda, \theta)) = \sum_{i=1}^N \log(P(X^{(i)}, Z^{(i)} | \lambda, \theta)) = \sum_{i=1}^N \sum_{k=1}^2 Z^{(ik)} \log(\lambda_k P(X^{(i)} | \theta_k))$$

$P(X^{(i)} | \theta_k)$ follows a Bernoulli distribution with probability θ for each position upto position L . So

$$P(X^{(i)} | \theta_k) = \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1 - X_l^{(i)})}$$

Substituting

$$\begin{aligned} \log(P(X, Z | \lambda, \theta)) &= \sum_{i=1}^N \sum_{k=1}^2 Z^{(ik)} \log \left(\lambda_k \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1 - X_l^{(i)})} \right) \\ &= \sum_{i=1}^N \sum_{k=1}^2 Z^{(ik)} \left(\log(\lambda_k) + \sum_{l=1}^L (X_l^{(i)} \log(\theta_k) + (1 - X_l^{(i)}) \log(1 - \theta_k)) \right) \\ &= \sum_{i=1}^N Z^{(i1)} \left(\log(\lambda_1) + \sum_{l=1}^L (X_l^{(i)} \log(\theta_1) + (1 - X_l^{(i)}) \log(1 - \theta_1)) \right) + Z^{(i2)} \left(\log(\lambda_2) + \sum_{l=1}^L (X_l^{(i)} \log(\theta_2) + (1 - X_l^{(i)}) \log(1 - \theta_2)) \right) \end{aligned}$$

(4b) The responsibility

Let us define the responsibility of component k for observation i as the posterior probability that $Z^{(ik)}$ is 1:

$$\gamma_{ik} := P(Z^{(ik)} = 1 | X^{(i)}, \lambda, \theta).$$

We express γ_{ik} in terms of $X^{(i)}$, λ , and θ .

$$\gamma_{ik} = P(Z^{(ik)} = 1 | X^{(i)}, \lambda, \theta)$$

Using Bayes theorem:

$$P(G|C) = \frac{P(G)P(C|G)}{P(C)}$$

Resulting in:

$$P(Z^{(ik)} = 1|X^{(i)}, \lambda, \theta) = \frac{P(Z^{(ik)} = 1|\lambda)P(X^{(i)}|Z^{(ik)} = 1, \theta)}{P(X^{(i)}|\lambda, \theta)}$$

Substituting the Model Components: Prior Probability:

$$P(Z^{(ik)} = 1|\lambda) = \lambda_k$$

Likelihood:

$$P(X^{(i)}|Z^{(ik)} = 1, \theta) = \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1-X_l^{(i)})}$$

Resulting in:

$$= \frac{\lambda_k \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1-X_l^{(i)})}}{P(X^{(i)}|\lambda, \theta)}$$

$P(X^{(i)}|\lambda, \theta)$ is the marginal probability of $X^{(i)}$, so one needs to sum over $Z^{(ik)} = 1$.

$$P(X^{(i)}|\lambda, \theta) = \sum_{j=1}^2 \lambda_j \prod_{l=1}^L \theta_j^{X_l^{(i)}} (1 - \theta_j)^{(1-X_l^{(i)})}$$

Resulting in:

$$\gamma_{ik} = \frac{\lambda_k \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1-X_l^{(i)})}}{\sum_{j=1}^2 \lambda_j \prod_{l=1}^L \theta_j^{X_l^{(i)}} (1 - \theta_j)^{(1-X_l^{(i)})}}$$

(4c) The M-step update formula

We derive the M-step update formulae for parameters λ and θ .

The M-step maximise $F(q, \theta)$ with respect to θ :

$$F(q, \theta) = E_q[l_{hid}(\theta)] - E_q[\log(q(Z))]$$

Since only $E_q[l_{hid}(\theta)]$ depends on θ , $E_q[l_{hid}(\theta)]$ is maximised. For a motif mixture model, as shown in the lectures, the complete data likelihood can be further simplified to:

$$E_q[l_{hid}(\theta)] = \sum_{i,k} \gamma_{ik} \log(\lambda_k) + \sum_{i,k} \gamma_{ik} \log(P(X^{(i)}|\theta_k))$$

Maximisation with respect to the λ_k involves only the left sum term.

$$J(\lambda, \mu) = \mu \left(1 - \sum_{k=1}^2 \lambda_k \right) + \sum_{i=1}^N \sum_{k=1}^2 \gamma_{ik} \log \lambda_k$$

where μ is a Lagrange multiplier for the constraint $\sum_{k=1}^2 \lambda_k = 1$. The update formula for λ_k is obtained by setting the derivative of this objective to zero:

$$\frac{\partial J}{\partial \lambda_k} = \frac{\sum_{i=1}^N \gamma_{ik}}{\lambda_k} - \mu = 0 \quad \text{for each } k, \quad \text{and} \quad \sum_{k=1}^2 \lambda_k = 1.$$

Rearranging, we have that

$$\sum_{i=1}^N \gamma_{ik} = \mu \lambda_k, \quad \text{for each } k.$$

Summing the two equations from $k = 1$ and $k = 2$ together, and using the fact that by definition $\sum_{k=1}^2 \gamma_{ik} = 1$ for each i and $\sum_{k=1}^2 \lambda_k = 1$, we have that $\mu = N$, and where N is the total number of cells.

$$\lambda_k^{new} = \frac{\sum_{i=1}^N \gamma_{ik}}{N} = \frac{\sum_{i=1}^N \frac{\lambda_k \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1 - X_l^{(i)})}}{\sum_{j=1}^2 \lambda_j \prod_{l=1}^L \theta_j^{X_l^{(i)}} (1 - \theta_j)^{(1 - X_l^{(i)})}}}{N}$$

Regarding maximising for θ_k , it is the right sum that should be maximised (because it is the sum that depends on θ_k). Since from (4a)

$$P(X^{(i)} | \theta_k) = \prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1 - X_l^{(i)})}$$

The right sum becomes

$$\begin{aligned} \sum_{i=1}^N \gamma_{ik} \log(P(X^{(i)} | \theta_k)) &= \sum_{i=1}^N \gamma_{ik} \log \left(\prod_{l=1}^L \theta_k^{X_l^{(i)}} (1 - \theta_k)^{(1 - X_l^{(i)})} \right) = \\ &= \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L (X_l^{(i)} \log(\theta_k) + (1 - X_l^{(i)}) \log(1 - \theta_k)) \end{aligned}$$

For ease the expression above is set $= f$. In order to find the optimal θ the first derivative is set $= 0$.

$$\begin{aligned} f &= \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L (X_l^{(i)} \log(\theta_k) + (1 - X_l^{(i)}) \log(1 - \theta_k)) = \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L (X_l^{(i)} \log(\theta_k) + (1 - X_l^{(i)}) \log(1 - \theta_k)) \\ \frac{\partial f}{\partial \theta_k} &= 0 = \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L \left(\frac{X_l^{(i)}}{\theta_k} - \frac{1 - X_l^{(i)}}{1 - \theta_k} \right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L \left(\frac{X_l^{(i)}(1-\theta_k) - \theta_k(1-X_l^{(i)})}{\theta_k(1-\theta_k)} \right) = \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L \left(\frac{X_l^{(i)} - \theta_k}{\theta_k(1-\theta_k)} \right) \\
&\Rightarrow \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L \left(\frac{\theta_k}{\theta_k(1-\theta_k)} \right) = \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L \left(\frac{X_l^{(i)}}{\theta_k(1-\theta_k)} \right) \\
&\Rightarrow \left(\frac{\theta_k}{\theta_k(1-\theta_k)} \right) \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L 1 = \left(\frac{1}{\theta_k(1-\theta_k)} \right) \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L X_l^{(i)} \\
&\Rightarrow \theta_k \cdot L \sum_{i=1}^N \gamma_{ik} = \sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L X_l^{(i)} \\
&\Rightarrow \theta_k^{new} = \frac{\sum_{i=1}^N \gamma_{ik} \sum_{l=1}^L X_l^{(i)}}{L \sum_{i=1}^N \gamma_{ik}}
\end{aligned}$$

(5) TSG inactivation problem implementation

(5a) The EM algorithm

Please write your code in the indicated blocks.

```

##' This function implements the EM algorithm for the gene mutation problem
##' @param D Data matrix of dimensions N-by-100, where N is the number of observations
##' @param k Number of cell groups
##' @param lambda Vector of mixture weights
##' @param theta Vector of probabilities of mutations
##' @param tolerance A threshold used to check convergence
mutation_EM <- function(D, k, lambda, theta, tolerance = 1e-2) {

  # observed log-likelihood
  ll_obs <- -Inf
  # difference between two iterations
  diff <- Inf
  # number of observations
  N <- nrow(D)
  # responsibilities
  gamma <- matrix(0, nrow = k, ncol = N)
  # keep track of lambda and theta during the optimization
  lambda_all <- lambda
  theta_all <- theta

```

```

# iteration number
t <- 1

# run the E-step and M-step until convergence
while (diff > tolerance) {

  ##### E-step #####

  ### YOUR CODE STARTS ###

  # Compute the responsibilities
  # ...

  # Update observed log-likelihood
  # ll_obs_new <- ...
  ll_obs <- c(ll_obs, ll_obs_new)
  # keep track of ll_obs throughout iterations

  # Recompute difference between two iterations
  diff <- abs(ll_obs[t] - ll_obs[t+1])

  ##### M-step #####

  # Recompute priors (mixture weights)
  # ...

  lambda_all <- rbind(lambda_all, lambda)
  # keep track of lambda throughout iterations

  # Recompute probability of mutation for each cell group
  # ...

  theta_all <- rbind(theta_all, theta)
  # keep track of theta throughout iterations

  t <- t+1
}
### YOUR CODE ENDS ###

return(list(ll_obs = ll_obs, lambda = lambda, theta = theta, gamma = gamma,

```

```

        lambda_all = lambda_all, theta_all = theta_all))
}

```

(5b) Load data, initialize parameters, and run the EM algorithm.

```

# read the data into D
D <- read.csv("mutations.csv")
D <- as.matrix(D)
# check the dimension of D
all(dim(D) == c(200, 100))

```

```
[1] TRUE
```

```

# number of cell groups
k <- 2
# mixture weights (a vector of length k)
lambda <- c(0.5, 0.5)
# probabilities of mutation (having a 1 in the data) (a vector of length k)
theta <- c(0.6, 0.4)

# run the EM algorithm
#res <- mutation_EM(D, k, lambda, theta)

```

(5c) Plotting convergence.

Observed log-likelihood throughout iterations:

```
## YOUR CODE ##
```

Probabilities of mutation throughout iterations:

```
## YOUR CODE ##
```

Mixture weights throughout iterations:

```
## YOUR CODE ##
```

(5d) Responsibilities at final iteration:

Heatmap of responsibilities:

```
## YOUR CODE ##
```

How many cells belong to the mutated group, and how many cells belong to the wild-type group?